

§ Lecture 15.0

Monday, 5 October 2026 17:29

What is missing?

$$\lim_{n \rightarrow \infty} a_n \stackrel{??}{=} \lim_{n \rightarrow \infty} a_n$$

Cauchy sequences of reals: let $(a_n)_{n \geq 1}$ be a sequence of real numbers. We say that $(a_n)_{n \geq 1}$ is a Cauchy sequence iff for all real $\varepsilon > 0$

$$\exists N \geq 1 \text{ s.t. } |a_m - a_n| \leq \varepsilon \quad \forall m, n \geq N.$$

Claim: let $(a_n)_{n \geq 1}$ be a sequence of rational numbers. Then the following two statements are equivalent.

$$(i) \quad \forall \text{ rational } \varepsilon > 0 \quad \exists N \geq 1 \text{ s.t. } |a_n - a_m| \leq \varepsilon \quad \forall m, n \geq N.$$

$$(ii) \quad \forall \text{ real } \varepsilon > 0 \quad \exists N \geq 1 \text{ s.t. } |a_n - a_m| \leq \varepsilon \quad \forall m, n \geq N.$$

Proof: $(i) \Rightarrow (ii)$

Fix a real $\varepsilon' > 0$. Then there exists a rational ε s.t. $0 < \varepsilon < \varepsilon'$.

$$\text{From (i)} \quad \exists N \geq 1 \text{ s.t. } \forall m, n \geq N \quad |a_n - a_m| \leq \varepsilon \leq \varepsilon'.$$

That is (ii) is true.

$(ii) \Rightarrow (i)$ let $\varepsilon' > 0$ be a rational number. Since it is also real.

$$\Rightarrow \exists N \geq 1 \text{ s.t. } |a_n - a_m| \leq \varepsilon' \quad \forall m, n \geq N.$$

Convergence of sequences. let $(a_n)_{n \geq 1}$ be a sequence of real numbers.

and let L be a real number. We say that $(a_n)_{n \geq 1}$ converges

to L iff $\forall \varepsilon > 0 \quad \exists N \geq 1 \text{ s.t.}$

$$|a_n - L| \leq \varepsilon \quad \forall n \geq N.$$

Lemma (Uniqueness of limits): let $(a_n)_{n \geq 1}$ be a sequence of real numbers. and $L \neq L'$ be two real numbers. Then it is not possible for $(a_n)_{n \geq 1}$ to converge to both L and L' .

Proof: Suppose $(a_n)_{n \geq 1}$ converges to both L and L' . That is

$$\forall \varepsilon_1 > 0 \quad \exists N_1 \geq 1 \text{ s.t. } |a_n - L| \leq \varepsilon_1 \quad \forall n \geq N_1 \quad \rightarrow (1)$$

$$\& \quad \forall \varepsilon_2 > 0 \quad \exists N_2 \geq 1 \text{ s.t. } |a_n - L'| \leq \varepsilon_2 \quad \forall n \geq N_2 \quad \rightarrow (2)$$

$$\text{Choose } \varepsilon_1 = \frac{|L - L'|}{3} = \varepsilon_2 > 0.$$

$$N = \max \{N_1, N_2\}.$$

Then $\forall n \geq N$

$$\begin{aligned} |L - L'| &= |L - a_n + a_n - L'| \\ &\leq |a_n - L| + |a_n - L'| \\ &\leq 2 \frac{|L - L'|}{3} \end{aligned}$$

This is a contradiction.

Thus a sequence can't converge to L and L' with $L \neq L'$.

Def (Limits of sequences):

if a sequence $(a_n)_{n \geq 1}$ of real numbers converges to some real number L , we say that $(a_n)_{n \geq 1}$ is convergent and its limit is L . That is,

$$L = \lim_{n \rightarrow \infty} a_n.$$

Otherwise it is called divergent and $\lim_{n \rightarrow \infty} a_n$ is undefined.

Remark: The limit doesn't care if the sequence starts from m or 1.

$$\lim_{n \rightarrow \infty} a_n, \quad \lim_{k \rightarrow \infty} a_k, \quad \text{etc. are all same.}$$

$$\text{Ex: } \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Proof: Fix real $\varepsilon > 0$. Then $\forall n \geq N$

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} \leq \frac{1}{N}$$

Given real $\varepsilon > 0$ we can pick N s.t. $N\varepsilon > 1$ or $\frac{1}{N} < \varepsilon$.

$$\Rightarrow |a_n - 0| \leq \varepsilon \quad \forall n \geq N.$$

Thus $(\frac{1}{n})_{n \geq 1}$ converges to 0.

Lemma (All convergent sequences are Cauchy): let $(a_n)_{n \geq 1}$ be a convergent sequence of reals. Then $(a_n)_{n \geq 1}$ is a Cauchy sequence.

Proof: let $(a_n)_{n \geq 1}$ converges to L . That is $\forall \varepsilon_1 > 0$

$$\exists N \geq 1 \text{ s.t. } |a_n - L| \leq \varepsilon_1 \quad \forall n \geq N. \quad \rightarrow (1)$$

let $\varepsilon > 0$ be a real number. Then apply eqⁿ (1) for $m, n \geq N$ and

$$\varepsilon_1 = \varepsilon/2. \text{ to get } \forall m, n \geq N$$

$$\begin{aligned} |a_n - a_m| &\leq |a_n - L| + |a_m - L| \\ &\leq \varepsilon \end{aligned}$$

$\Rightarrow (a_n)_{n \geq 1}$ is a Cauchy sequence.

§ Lecture 15.1

Monday, 5 October 2026 18:04

Claim: Suppose $(a_n)_{n \geq 1}$ is a Cauchy sequence of rationals. Then $(a_n)_{n \geq 1}$

converges to $\lim_{n \rightarrow \infty} a_n$, i.e.

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_n.$$

Proof: Since $(a_n)_{n \geq 1}$ is a Cauchy sequence of rationals, then $\lim_{n \rightarrow \infty} a_n$ is a real number.

Further $\forall \epsilon > 0 \exists N \geq 1$ s.t. $|a_n - a_m| \leq \epsilon \forall n, m \geq N$.

In particular fix $m \geq N$. Then

$$a_n - \epsilon \leq a_n \leq a_m + \epsilon \quad \forall n \geq N.$$

Define $b_k = a_{n+k-1}$ ($b_1 = a_n, b_2 = a_{n+1}, \dots$)

Then

$$a_n - \epsilon \leq b_k \leq a_n + \epsilon \quad \forall k \geq 1.$$

$$\Rightarrow a_n - \epsilon \leq \lim_{k \rightarrow \infty} b_k \leq a_n + \epsilon$$

But $(b_k)_{k \geq 1}$ and $(a_n)_{n \geq 1}$ are equivalent.

$$\Rightarrow |a_n - \lim_{n \rightarrow \infty} a_n| \leq \epsilon$$

Thus for all $\epsilon > 0, \exists N \geq 1$ s.t. $|a_n - \lim_{n \rightarrow \infty} a_n| \leq \epsilon \quad \forall n \geq N$.

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} a_n.$$

Def. (Bounded sequences). We say that a sequence $(a_n)_{n \geq 1}$ of reals is

bounded iff $\exists$ a real number $M > 0$ s.t.

$$|a_n| \leq M \quad \forall n \geq 1.$$

Claim: Every convergent sequence is bounded.

Proof: Let $(a_n)_{n \geq 1}$ be a convergent sequence of reals, converging to real number L .

i.e. $\forall \epsilon > 0 \exists N \geq 1$ s.t. $|a_n - L| \leq \epsilon \quad \forall n \geq N$.

Choosing $\epsilon = 1$

$$|a_n| = |a_n - L + L| \leq |a_n - L| + |L| \leq 1 + |L| \quad \forall n \geq N.$$

Since $(a_1, \dots, a_{N-1})$ is a finite sequence, it is bounded, let $M_0 > 0$ be

$$\text{s.t.} \quad |a_i| \leq M_0 \quad \forall 1 \leq i \leq N-1.$$

Then $|a_n| \leq M \quad \forall n \geq 1$, where $M = \max\{M_0, |L| + 1\} > 0$.

Limit laws:- Let $(a_n)_{n \geq 1}$ and $(b_n)_{n \geq 1}$ be convergent sequences of

reals. Let

$$x = \lim_{n \rightarrow \infty} a_n, \quad y = \lim_{n \rightarrow \infty} b_n.$$

Then

$$(i) \quad \lim_{n \rightarrow \infty} (a_n \pm b_n) = x \pm y = \lim_{n \rightarrow \infty} a_n \pm \lim_{n \rightarrow \infty} b_n$$

$$(ii) \quad \lim_{n \rightarrow \infty} (a_n b_n) = xy = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right)$$

(iii) Suppose $y \neq 0$, and $b_n \neq 0 \quad \forall n \geq 1$. Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}$$

$$(iv) \quad \lim_{n \rightarrow \infty} \max/\min(a_n, b_n) = \max/\min(\lim_{n \rightarrow \infty} a_n, \lim_{n \rightarrow \infty} b_n).$$

Proof: Given

$$\forall \epsilon_1 > 0 \exists N_1 \geq 1 \text{ s.t. } \forall n \geq N_1 \quad |a_n - x| \leq \epsilon_1. \rightarrow (1)$$

$$\forall \epsilon_2 > 0 \exists N_2 \geq 1 \text{ s.t. } \forall n \geq N_2 \quad |b_n - y| \leq \epsilon_2 \rightarrow (2)$$

Pick $N = \max\{N_1, N_2\}$. Choose $\epsilon > 0$.

Then $\forall n \geq N$

$$|a_n \pm b_n - (x \pm y)| \leq |a_n - x| + |b_n - y| \leq \epsilon_1 + \epsilon_2$$

Pick $\epsilon_1 = \epsilon_2 = \frac{\epsilon}{2}$ to get

$$|a_n \pm b_n - (x \pm y)| \leq \epsilon \quad \forall n \geq N.$$

$$\Rightarrow \lim_{n \rightarrow \infty} (a_n \pm b_n) = x \pm y.$$

(ii) Fix $\epsilon > 0$. $\forall n \geq N$

$$|a_n b_n - xy| \leq |a_n - x| |b_n| + |b_n - y| |x|$$

$$\leq M_1 \epsilon_1 + |x| \epsilon_2 \quad (\text{Convergent sequence are bounded})$$

$$\text{Choose } \epsilon_1 = \frac{\epsilon}{2M_1}, \quad \epsilon_2 = \frac{\epsilon}{2(|x|+1)}$$

$$\Rightarrow |a_n b_n - xy| \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} \frac{|x|}{|x|+1} \leq \epsilon.$$

$$\Rightarrow \lim_{n \rightarrow \infty} (a_n b_n) = xy.$$

(iii) Given $b_n \neq 0 \quad \forall n \geq 1$, $y \neq 0$. Then

$$(b_n^{-1}) \text{ is a convergent sequence converging to } y^{-1}.$$

Then again just like $(a_n b_n)$, we find $(a_n b_n^{-1})$ converges to xy^{-1} .

(iv) Note that

$$\max(a, b) = \frac{a+b+|a-b|}{2}$$

$$\min(a, b) = \frac{a+b-|a-b|}{2}$$

$$\text{Further, if } \lim_{n \rightarrow \infty} a_n = x \Rightarrow \lim_{n \rightarrow \infty} |a_n| = |x|.$$

Proof: From triangle inequality:

$$|a_n| \leq |a_n - x| + |x| \Rightarrow |a_n| - |x| \leq |a_n - x|$$

$$|x| \leq |a_n - x| + |a_n| \Rightarrow |a_n| - |x| \geq -|a_n - x|$$

$$\Rightarrow ||a_n| - |x|| \leq |a_n - x|$$

$$\text{Then since } \forall \epsilon_1 > 0 \exists N_1 \geq 1 \text{ s.t. } |a_n - x| \leq \epsilon_1 \quad \forall n \geq N_1,$$

$$\text{we have } ||a_n| - |x|| \leq |a_n - x| \leq \epsilon_1 \quad \forall n \geq N_1$$

$$\lim_{n \rightarrow \infty} |a_n| = |x|.$$

$$\Rightarrow \forall \epsilon_3 > 0, \exists N_3 \geq 1 \text{ s.t. } ||a_n - b_n| - |x - y|| \leq \epsilon_3 \quad \forall n \geq N_3$$

Now

$$|\max(a_n, b_n) - \max(x, y)| = \left| \frac{a_n + b_n + |a_n - b_n|}{2} - \frac{x + y + |x - y|}{2} \right|$$

$$\leq \frac{1}{2} |a_n - x| + \frac{1}{2} |b_n - y| + \frac{1}{2} ||a_n - b_n| - |x - y||$$

$$\leq \frac{\epsilon_1 + \epsilon_2 + \epsilon_3}{2} \quad \forall n \geq \max\{N_1, N_2, N_3\}$$

$$\text{Choose } \epsilon_1 = 2\epsilon_3, \quad \epsilon_2 = 2\epsilon_3, \quad \epsilon_3 = 2\epsilon_3$$

$$\Rightarrow \lim_{n \rightarrow \infty} \max(a_n, b_n) = \max(x, y).$$

Similarly for min.